

Quantum Cryptography Based on Bells Theorem

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1. ABSTRACT & THEORETICAL FOUNDATION

We formulate entanglement-based quantum cryptography, showing that violation of the Clauser-Horne-Shimony-Holt (CHSH) Bell inequality guarantees the secrecy of distributed cryptographic keys against arbitrary eavesdroppers.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Instead of sending keys directly, a central source sends pairs of entangled twins to Alice and Bob. If a spy tampers with either twin, the entanglement breaks, exposing the spy immediately."

3. Mathematical Formulation & Unitary Rule

$$S = E(a_1, b_1) - E(a_1, b_3) + E(a_3, b_1) + E(a_3, b_3) = -2\sqrt{2}, \quad \text{quad } |S| \leq 2$$

Violation of Bell/CHSH inequality confirms entanglement and proves zero third-party information leakage.

4. Step-by-Step Circuit Sequence

Bell state generation (H + CNOT), distribution to Alice and Bob, randomized directional measurement.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2, 2)
# Create Bell singlet state |Psi->
qc.x([0, 1])
qc.h(0)
qc.cx(0, 1)

# Alice measures qubit 0, Bob measures qubit 1
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Bounded by Bell-CHSH inequality $|S| \leq 2$ reaches Tsirelson's bound $|S| = 2\sqrt{2}$ Foundational

7. Key Applications & Future Research Directions

- Device-Independent Quantum Key Distribution (DI-QKD)
- Certified random number generation verified by Bell tests
- Space-based satellite entanglement distribution networks

Future Work:

Quantum repeaters, entanglement purification networks, and multi-party quantum secret sharing.